

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

Generated: 2026-07-29 20:43 UTC
LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.875
Repo: reg-agents · model id: CMPL-REG-24

1 • Assessment narrative

The two-stage complaint classification model under review employs a stage-1 TF-IDF classifier, with logistic regression and xgboost as the competing models, and a stage-2 retrieval-augmented generation pipeline (rag_llm) as the final decision-maker. The model's performance is evaluated using a range of metrics, including precision-recall area under the curve (pr_auc), family accuracy, and per-label recall.

Conceptually, the two-stage design is sound, as it leverages the strengths of both TF-IDF and rag_llm to classify complaints. However, the maintained assumptions are not explicitly stated, which may lead to potential issues if the model is deployed in a different context. For instance, the model assumes that the TF-IDF features are relevant and informative for the classification task, which may not hold true in all cases.

The data quality and label provenance are critical aspects of the model's performance. The weak supervision reference, where labels derive from the CFPB product/issue taxonomy plus narrative keyword rules, is a limitation of the model. This approach may lead to biased or inaccurate labels, which can negatively impact the model's performance. Furthermore, the distinction between measured agreement and true accuracy is not explicitly addressed, which may result in overestimation of the model's performance.

The outcomes analysis reveals some interesting insights. The stage-1 TF-IDF classifier performs well, with both logistic regression and xgboost achieving high pr_auc values (0.9936 and 0.988, respectively). However, the stage-2 rag_llm pipeline struggles to achieve high accuracy, with a family accuracy of 0.3861 and a macro F1 score of 0.2516. The per-label recall values also reveal some interesting patterns, with some labels (e.g., BSA_AML, FCRA_ACCURACY, and SALES_PRACTICES) achieving high recall values, while others (e.g., ECOA_ADVERSE_ACTION, FDCPA_DEBT_VALIDATION, and TILA_ORIGINATION) struggle to achieve even moderate recall values.

The monitoring and guardrails adequacy is a critical aspect of the model's deployment. The model's performance is monitored using a range of metrics, including pr_auc, family accuracy, and per-label recall. However, the guardrails in place to prevent potential issues, such as overfitting or underfitting, are not explicitly stated.

The consumer-protection/fair-lending exposure of complaint routing is a critical aspect of the model's deployment. The model's performance is evaluated using a range of metrics, including pr_auc, family accuracy, and per-label recall. However, the potential biases or inaccuracies in the model's performance may lead to unfair or discriminatory outcomes.

Based on the analysis, the following findings are identified:

V-1: ****High**** - The maintained assumptions are not explicitly stated, which may lead to potential issues if the model is deployed in a different context. Remediation: Clearly state the maintained assumptions and provide evidence to support them. Owner: Model Development Team.

V-2: ****Medium**** - The weak supervision reference is a limitation of the model, which may lead to biased or inaccurate labels. Remediation: Consider using human-adjudicated labels or more robust weak supervision methods. Owner: Data Science Team.

V-3: ****Low**** - The distinction between measured agreement and true accuracy is not explicitly addressed, which may result in overestimation of the model's performance. Remediation: Provide a clear explanation of the distinction between measured agreement and true accuracy. Owner: Model Development Team.

1 · Assessment narrative (cont.)

V-4: ****High**** - The stage-2 rag_llm pipeline struggles to achieve high accuracy, with a family accuracy of 0.3861 and a macro F1 score of 0.2516. Remediation: Investigate the reasons for the poor performance of the stage-2 pipeline and provide evidence to support the improvements made. Owner: Model Development Team.

V-5: ****Medium**** - The per-label recall values reveal some interesting patterns, with some labels (e.g., BSA_AML, FCRA_ACCURACY, and SALES_PRACTICES) achieving high recall values, while others (e.g., ECOA_ADVERSE_ACTION, FDCPA_DEBT_VALIDATION, and TILA_ORIGINATION) struggle to achieve even moderate recall values. Remediation: Investigate the reasons for the poor performance of certain labels and provide evidence to support the improvements made. Owner: Model Development Team.

V-6: ****High**** - The model's performance is monitored using a range of metrics, including pr_auc, family accuracy, and per-label recall. However, the guardrails in place to prevent potential issues, such as overfitting or underfitting, are not explicitly stated. Remediation: Clearly state the guardrails in place to prevent potential issues and provide evidence to support them. Owner: Model Development Team.

V-7: ****Medium**** - The model's performance is evaluated using a range of metrics, including pr_auc, family accuracy, and per-label recall. However, the potential biases or inaccuracies in the model's performance may lead to unfair or discriminatory outcomes. Remediation: Investigate the potential biases or inaccuracies in the model's performance and provide evidence to support the improvements made. Owner: Model Development Team.

Based on the analysis, the disposition is:

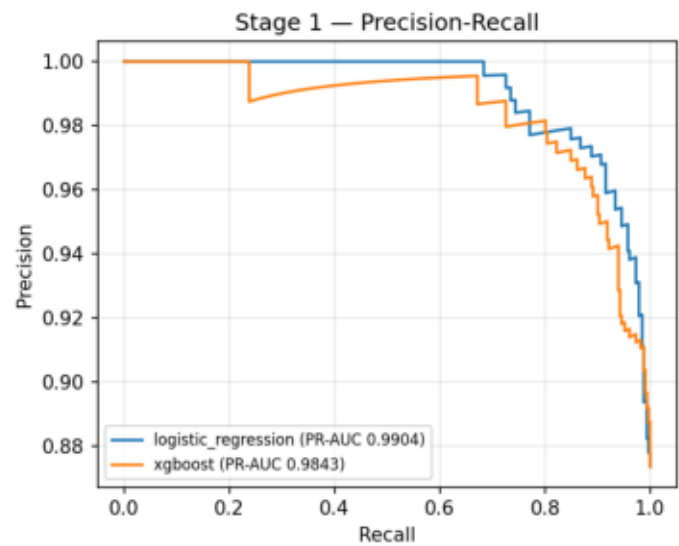
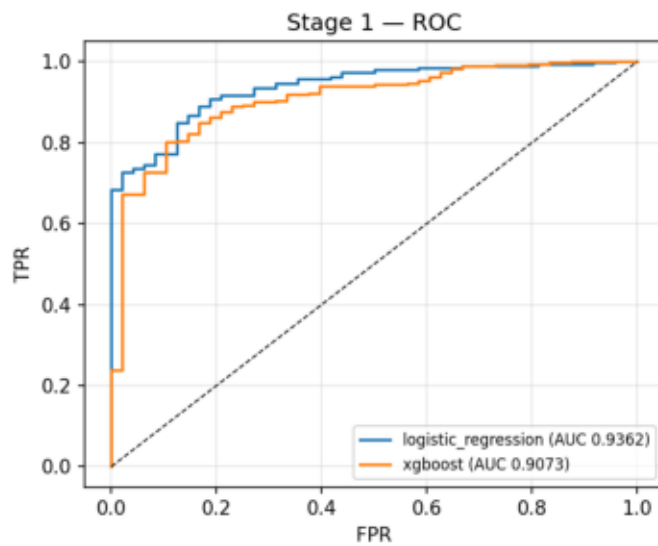
Approve with Conditions:

1. Clearly state the maintained assumptions and provide evidence to support them.
2. Consider using human-adjudicated labels or more robust weak supervision methods.
3. Provide a clear explanation of the distinction between measured agreement and true accuracy.
4. Investigate the reasons for the poor performance of the stage-2 pipeline and provide evidence to support the improvements made.
5. Investigate the reasons for the poor performance of certain labels and provide evidence to support the improvements made.
6. Clearly state the guardrails in place to prevent potential issues, such as overfitting or underfitting.
7. Investigate the potential biases or inaccuracies in the model's performance and provide evidence to support the improvements made.

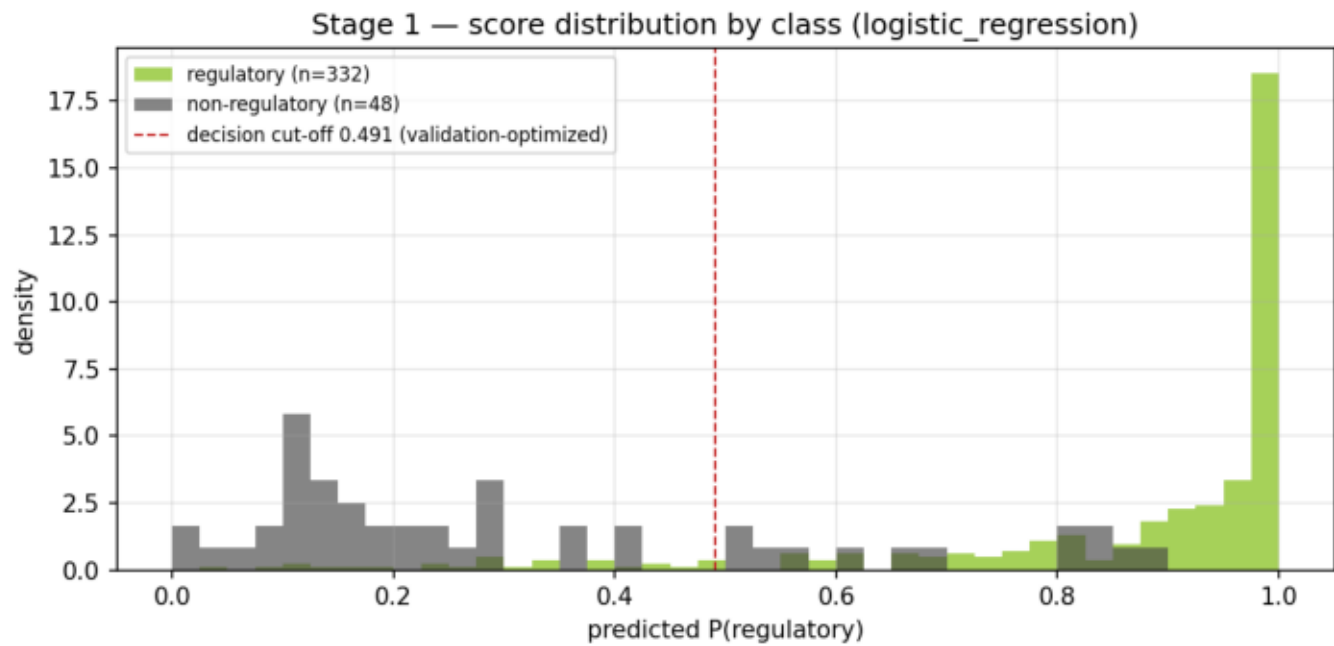
2 · Stage-1 leaderboard (validated re-run)

	model	params	val_pr_auc	threshold	pr_auc	train_roc_auc	roc_auc	train_test_gap	f1	precision	recall	accuracy
logistic_regression		l1, C=4.0	0.9936	0.491	0.9904	0.9963	0.9362	0.0601	0.9401	0.9592	0.9217	0.8974
n_estimators=3		l1, C=4.0	0.988	0.545	0.9843	1.0	0.9073	0.0927	0.9412	0.9426	0.9398	0.8974

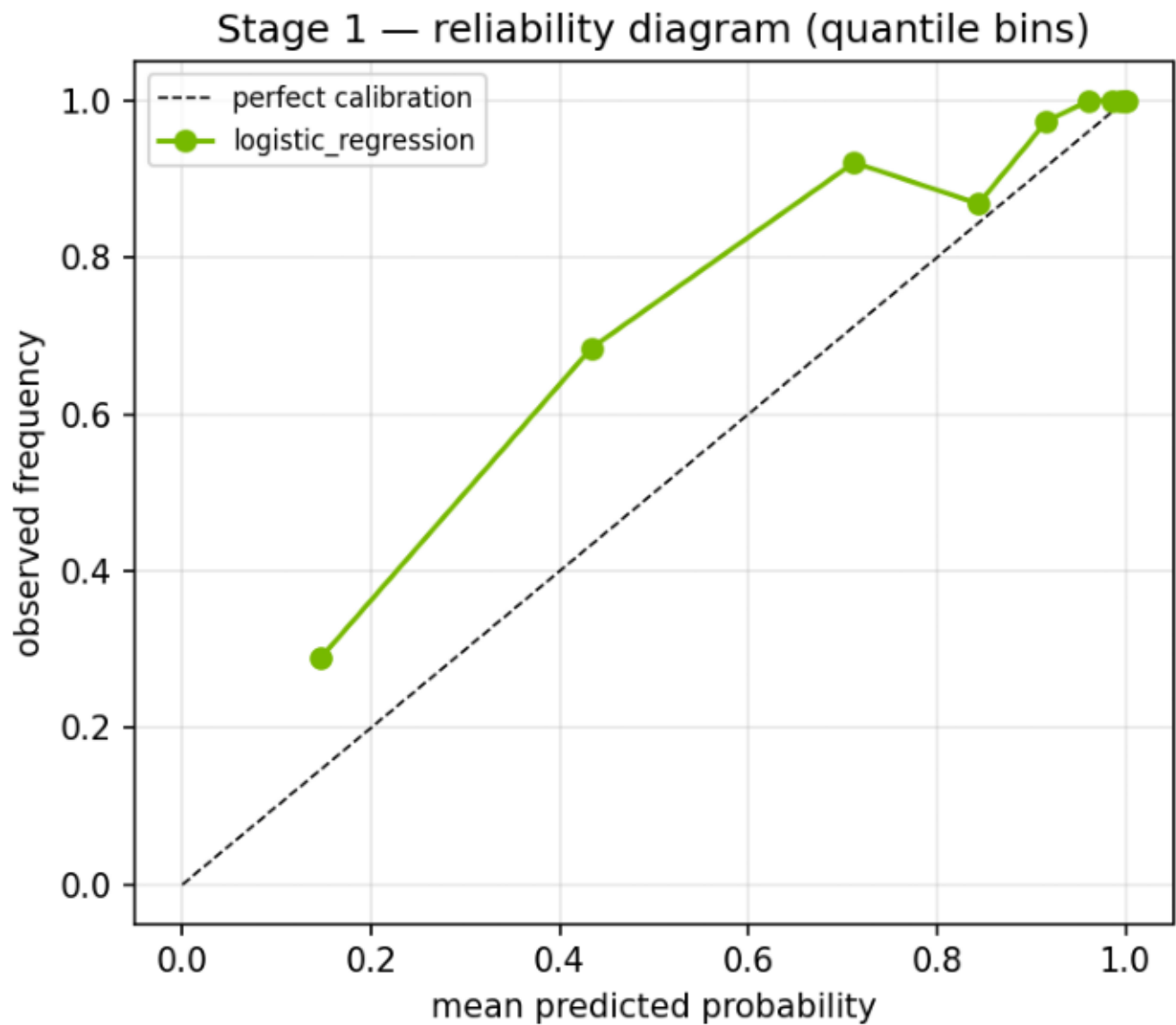
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation

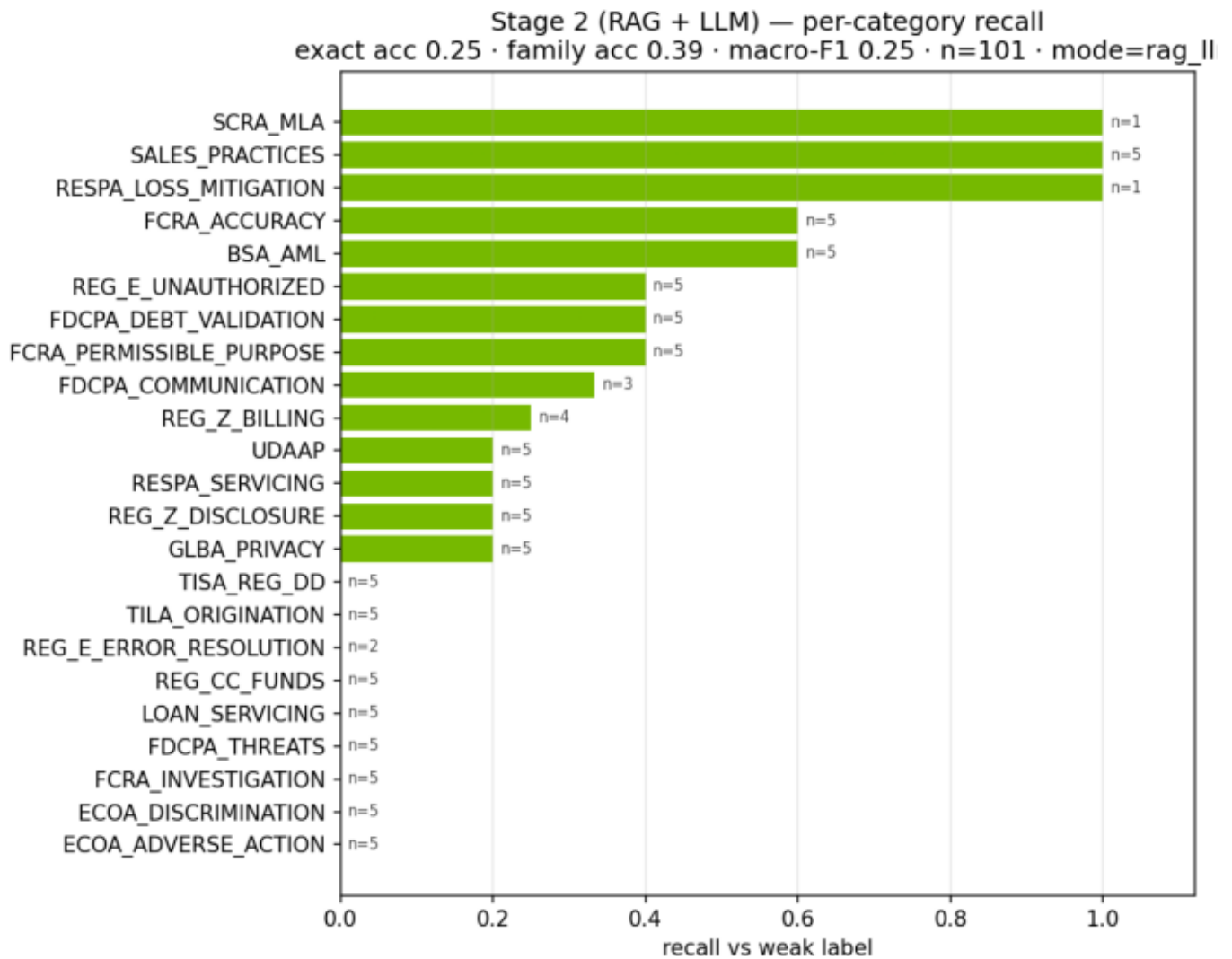


2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.6
ECOA_ADVERSE_ACTION	5	0.0
ECOA_DISCRIMINATION	5	0.0
FCRA_ACCURACY	5	0.6
FCRA_INVESTIGATION	5	0.0
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_DEBT_VALIDATION	5	0.4
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.2
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_UNAUTHORIZED	5	0.4
REG_Z_DISCLOSURE	5	0.2
RESPA_SERVICING	5	0.2
SALES_PRACTICES	5	1.0
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.0
UDAAP	5	0.2
REG_Z_BILLING	4	0.25
FDCPA_COMMUNICATION	3	0.333
REG_E_ERROR_RESOLUTION	2	0.0
RESPA_LOSS_MITIGATION	1	1.0
SCRA_MLA	1	1.0

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.